AATIF KHAN PATHAN

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of experience building production-grade deep learning systems for computer vision, NLP, and intelligent automation. Proven track record deploying 10+ ML models to GCP Vertex AI with automated CI/CD pipelines. Expertise in PyTorch, YOLO, BERT, reinforcement learning, and MLOps best practices. Achieved 99.58% image classification accuracy, 92.3% F1 for NER, and 70.6% mAP for real-time object detection. Specialized in end-to-end ML system architecture from research to production deployment with monitoring and drift detection. Currently on GCP team developing cloud-native ML solutions.

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch • TensorFlow • Scikit-learn • Neural Networks (CNN, RNN, LSTM, Transformers) • Computer Vision • Natural Language Processing • Reinforcement Learning (Q-Learning, DQN, Policy Gradients) • Transfer Learning • Model Optimization • Hyperparameter Tuning

Computer Vision: YOLO (YOLOv8) • OpenCV • MediaPipe • Object Detection • Image Classification • Real-time Video Processing • ResNet • EfficientNet • Facial Landmark Detection

NLP & LLMs: BERT • Transformers • Hugging Face • LangChain • RAG (Retrieval-Augmented Generation) • Vector Databases (ChromaDB) • Named Entity Recognition • Sentiment Analysis • Fine-tuning • Prompt Engineering • Gemini API • GPT

MLOps & DevOps: MLflow • Docker • CI/CD (GitHub Actions) • Model Versioning • Experiment Tracking • Model Monitoring • Drift Detection • FastAPI • REST APIs • Git/GitLab • Production Deployment

Cloud & Infrastructure: Google Cloud Platform (GCP Vertex AI, Cloud Storage, Cloud Build, Cloud Monitoring) • Serverless Deployment • Cloud ML Pipelines • Cost Optimization

Programming & Tools: Python (Advanced) • NumPy • Pandas • Matplotlib • Streamlit • Gradio • Jupyter • C++ • MATLAB/Simulink • SQL • Linux • VS Code

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Tata Consultancy Services

Gandhinagar, Gujarat | October 2022 – Present

- Architected and deployed production-grade computer vision systems for automotive applications using PyTorch CNNs, YOLO, OpenCV, and MediaPipe, serving real-time inference for driver monitoring and cabin intelligence across 1000+ vehicles
- Designed custom deep learning pipelines for driver safety features (distraction detection, drowsiness monitoring, yawn detection) achieving 30% improvement in model robustness through advanced data augmentation, focal loss optimization, and ensemble techniques
- Built production Retrieval-Augmented Generation (RAG) system using LangChain, vector databases (ChromaDB), and LLMs for document intelligence, enabling semantic search across 10,000+ technical documents and reducing information retrieval time by 60%
- Implemented LLM-powered solutions for engineering workflows using GPT API, BERT, and custom fine-tuned models for requirement summarization and technical knowledge extraction, processing 500+ documents monthly
- Developed automated ML workflows and control system models using MATLAB/Simulink, improving development efficiency by 25% through script-based automation and systematic testing procedures
- Collaborated cross-functionally with data engineers, product teams, and QA to deploy ML models meeting AUTOSAR compliance standards, conducting rigorous MIL/SIL/B2B testing for safety-critical automotive systems

Deep Learning Intern | CipherSchools

Jodhpur, Rajasthan | June 2021 – July 2021

- Built facial expression recognition CNN using PyTorch achieving 92% accuracy, enabling real-time emotion-based music recommendation system with sub-100ms inference latency
- Developed custom object detection model from scratch using PyTorch with 90%+ mAP for user-defined object classes, implementing data augmentation and transfer learning techniques
- Created NLP sentiment analysis pipeline processing 50,000+ tweets using BERT embeddings and classification models, achieving 88% F1-score on airline customer feedback data
- Engineered spam classification system using traditional ML and deep learning approaches, optimizing for 95% accuracy with minimal false positives through precision-recall curve analysis

ML BOOTCAMP PROJECTS (10 Production Systems - 28 Days)

1. Custom YOLO Object Detection System | *PyTorch, YOLOv8M, GCP Vertex AI, FastAPI, Docker*

- Trained custom YOLOv8M model (25.9M parameters) achieving 70.6% mAP@0.5 and 59% mAP@0.5:0.95 on proprietary dataset with 50 epochs, batch size 16, image size 640
- Engineered real-time detection system processing 20-25 FPS on CPU with webcam integration and video file processing capabilities
- Deployed to GCP Vertex AI with automated CI/CD pipeline using GitHub Actions, reducing deployment time from 2 hours (manual) to 8 minutes (automated)
- Implemented FastAPI REST endpoint for production inference with average latency of 145ms and P95 latency of 234ms

2. BERT Named Entity Recognition + Multi-Task NLP API | *BERT, Transformers, Hugging Face, FastAPI, GCP*

- Fine-tuned bert-base-cased (110M parameters) on CoNLL-2003 dataset achieving 92.3% F1 score, 91.9% precision, and 92.8% recall for named entity recognition
- Built multi-task NLP API combining NER, sentiment analysis (DistilBERT), topic classification (BART zero-shot), and question answering (DistilBERT-SQuAD) in unified FastAPI service
- Deployed to GCP Vertex AI with automated versioning, monitoring, and drift detection using Kolmogorov-Smirnov statistical tests
- Implemented production monitoring system logging 1000+ predictions daily with latency tracking and automated alerts for anomalies

3. LSTM + Prophet Time Series Forecasting System | *PyTorch, Prophet, Streamlit, Yahoo Finance API*

- Implemented LSTM model (50 hidden units, 2 layers, 60-day lookback) achieving ~2% MAPE for stock price forecasting with MinMaxScaler normalization
- Developed Prophet baseline model for comparison, achieving 30x faster training time with ~3-6% MAPE for automated trend and seasonality detection
- Created interactive Streamlit dashboard with real-time predictions, model comparison, and 30-day forecasting with confidence intervals

4. Reinforcement Learning: Q-Learning to Policy Gradients | *PyTorch, Gymnasium, Q-Learning, DQN, REINFORCE*

- Implemented tabular Q-Learning from scratch achieving 100% success rate on FrozenLake 4x4 environment with ε-greedy exploration and Bellman equation updates
- Built Deep Q-Network (DQN) with experience replay (10,000 buffer) and target networks solving CartPole with 215 average reward over 100 episodes
- Developed REINFORCE policy gradient algorithm for continuous action spaces achieving 200+ average reward on CartPole, establishing foundation for robotics applications

5. Production RAG System for Technical Documentation | *LangChain, ChromaDB, Gemini 2.5 Flash, Sentence Transformers*

- Architected retrieval-augmented generation pipeline processing 100-150 markdown document chunks with 500-character chunking and 50-character overlap

- Integrated Gemini 2.5 Flash LLM for generation achieving 2-5 second response times with 95% accuracy improvement over base LLM queries
- Implemented semantic search using sentence-transformers embeddings with ChromaDB vector database for efficient document retrieval

6. Driver Drowsiness Detection (Edge Deployment) | *EfficientNet-B0, OpenCV, PyTorch, Raspberry Pi*

- Developed real-time drowsiness detection system using EfficientNet-B0 achieving 93% accuracy on 3-class classification (alert, drowsy, sleeping)
- Optimized for edge deployment on Raspberry Pi through model quantization and efficient architecture selection, balancing accuracy and inference speed

7. Electrical Equipment Image Classifier | *ResNet50, PyTorch, Transfer Learning*

- Applied transfer learning with ResNet50 progressing from feature extraction (95.83% accuracy) to fine-tuning (99.58% accuracy) on custom electrical equipment dataset
- Optimized training with AdamW optimizer, learning rate scheduling, and strategic layer unfreezing for maximum performance

8. RNN/LSTM Text Generation | *LSTM, PyTorch, AdamW Optimization*

- Built character-level LSTM text generator achieving 3.75x faster convergence (loss 1.188→0.59) through AdamW optimization and learning rate scheduling
- Implemented temperature-based sampling and hidden state management for controllable text generation quality

9. LLM Fine-Tuning with LoRA/QLoRA | *LoRA, QLoRA, Hugging Face, PEFT*

- Mastered parameter-efficient fine-tuning achieving 99% parameter reduction through LoRA (Low-Rank Adaptation) low-rank decomposition
- Explored QLoRA 4-bit quantization reducing memory requirements from 14GB to 4GB while maintaining model performance

10. End-to-End ML Pipeline with MLOps Automation | *MLflow, FastAPI, Docker, GitHub Actions, GCP Vertex AI*

- Architected complete MLOps pipeline: data ingestion → validation → training (MLflow) → evaluation → model registry → deployment (FastAPI) → monitoring
- Implemented CI/CD automation with GitHub Actions achieving zero-downtime deployments, automated testing, and health checks
- Deployed to GCP Vertex AI with Cloud Monitoring for prediction logging, latency tracking (P50/P95/P99 metrics), and automated drift detection
- Managed 15+ model versions with MLflow registry enabling A/B testing, staged deployment (Staging→Production), and easy rollback capabilities

ADDITIONAL PROJECTS

AI-Powered Financial News Analysis & Trading Insights | *PyTorch, LLMs, Web Scraping, Gradio*

- Engineered automated market intelligence system scraping 50+ daily financial articles using BeautifulSoup with robust error handling
- Integrated Gemini LLM API generating concise summaries and actionable trading signals, reducing manual analysis time by 60%

Real-time Driver Safety System (Multi-modal) | *YOLO, MediaPipe, OpenCV*

- Developed multi-modal safety system combining YOLO phone detection and MediaPipe facial landmarks achieving 30+ FPS real-time performance
- Implemented eye aspect ratio (EAR), mouth opening ratio, and head pose estimation with temporal smoothing, reducing false positives by 40%

Handwritten Digit Recognition Web App | *PyTorch, CNN, Streamlit*

- Built end-to-end digit recognition using LeNet CNN achieving 98%+ accuracy on MNIST with production-ready Streamlit deployment

EDUCATION

Bachelor of Technology in Electrical Engineering | Bikaner Technical University

Jodhpur, Rajasthan | July 2018 – June 2022

CERTIFICATIONS & ACHIEVEMENTS

- Completed intensive 12-week ML Engineering Bootcamp: 10 production projects deployed to GCP Vertex AI with full MLOps automation
- LeetCode: 45+ problems solved across key patterns (Hash Maps, Two Pointers, Stack, Sliding Window, Trees, Binary Search, Dynamic Programming, Graphs)
- Built professional portfolio website showcasing all projects with quantified results and live demos
- Achieved exceptional model performance: 99.58% image classification, 92.3% F1 NER, 70.6% mAP object detection, 100% RL FrozenLake
- Deployed production ML systems reducing deployment time by 94% (2 hours→8 minutes) through CI/CD automation